

Лабораторная работа №6

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- **Группа:** 6204-010302D
- **Тема:** Поток и семафоры — генерация и интегрирование заданий

1. Цель работы:

- **Цель:** Разработать приложение, в котором один поток формирует задания для численного интегрирования, а второй поток их решает; исследовать и реализовать две схемы синхронизации (блокировки и одноместный семафор), обеспечить корректную обработку прерываний и корректность результатов.

2. Краткое описание реализации:

- `Functions.integrate`: метод численного интегрирования методом трапеций. Проверяет корректность аргументов (не *null*, не *NaN*, *step > 0*) и принадлежность интервала области определения функции; учитывает последний частичный сегмент.
- `threads.Task`: объект задания содержит ссылку на *Function*, *left*, *right*, *step*, поле *count*. Реализованы блокирующие методы *putBlocking/takeBlocking* (с *wait/notify*) и неблокирующие *putNoSync/takeNoSync*. Внутри есть вложенный *TaskSnapshot*.
- `threads.SimpleGenerator` / `threads.SimpleIntegrator`: реализации *Runnable* для простой версии с использованием блокирующих методов *Task*. Генератор пишет задачу, интегратор читает и вычисляет интеграл.
- `threads.SimpleSemaphore`: одноместный семафор, различающий операции чтения и записи (*acquireWriter/releaseWriter*, *acquireReader/releaseReader*).
- `threads.Generator` / `threads.Integrator`: классы *Thread*, использующие *SimpleSemaphore* и неблокирующие методы *Task* для семафорной версии. Обработывают *InterruptedException* корректно (устанавливают флаг прерывания и завершают цикл).
- `Main`: содержит *demoLab6()* — тесты для всех задач: интеграл *exp*, *nonThread()*, *simpleThreads()*, *complicatedThreads()*; в *complicatedThreads()* основной поток ждёт 50 ms, затем вызывает *interrupt()* для потоков.

3. Выполнение заданий:

- **Задание 1 — Интегрирование**
 - **Реализация:** `Functions.integrate(Function f, double left, double right,`

double step) – трапеций; обрабатывается частичный последний сегмент; выбрасывается *IllegalArgumentException* при выходе за область определения или некорректных аргументах.

- **Проверка:** вычислен точный интеграл экспоненты на $[0,1]$ и подобран шаг, при котором ошибка $< 5e-8$.

```
public static double integrate(Function f, double left, double right, double step) {
    if (f == null) throw new IllegalArgumentException(s: "Function is null");
    if (Double.isNaN(left) || Double.isNaN(right) || Double.isNaN(step))
        throw new IllegalArgumentException(s: "Arguments must be numbers");
    if (step <= 0) throw new IllegalArgumentException(s: "Step must be positive");

    double domainLeft = f.getLeftDomainBorder();
    double domainRight = f.getRightDomainBorder();
    if (left < domainLeft || right > domainRight) {
        throw new IllegalArgumentException(s: "Integration interval is outside function domain");
    }

    double a = Math.min(left, right);
    double b = Math.max(left, right);

    double sum = 0.0;
    double x = a;
    double fx = f.getFunctionValue(x);

    if (Double.isNaN(fx)) throw new IllegalArgumentException(s: "Function undefined at left border");

    while (x + step < b) {
        double xNext = x + step;
        double fxNext = f.getFunctionValue(xNext);
        if (Double.isNaN(fxNext)) throw new IllegalArgumentException(s: "Function undefined inside interval");
        sum += 0.5 * (fx + fxNext) * (xNext - x);
        x = xNext;
        fx = fxNext;
    }

    if (x < b) {
        double xNext = b;
        double fxNext = f.getFunctionValue(xNext);
        if (Double.isNaN(fxNext)) throw new IllegalArgumentException(s: "Function undefined at right border");
        sum += 0.5 * (fx + fxNext) * (xNext - x);
    }

    return left <= right ? sum : -sum;
}
```

- **Задание 2** — *nonThread()*

- **Реализация:** последовательная генерация count задач (минимум 100): случайная логарифмическая функция (основание $\in [1,10]$), $left \in [0,100]$, $right \in [100,200]$, $step \in [0,1]$; печать *Source* и *Result*.
- **Проверка:** вывод соответствующих строк в консоль; интеграция производится последовательно.

```
public synchronized void putBlocking(Function function, double left, double right, double step) throws InterruptedException {
    while (!available) {
        wait();
    }
    this.function = function;
    this.left = left;
    this.right = right;
    this.step = step;
    available = true;
    notifyAll();
}

public synchronized TaskSnapshot takeBlocking() throws InterruptedException {
    while (!available) {
        wait();
    }
    TaskSnapshot snap = new TaskSnapshot(function, left, right, step);
    available = false;
    function = null;
    notifyAll();
    return snap;
}
```

- **Задание 3** — *simpleThreads()*

- **Реализация:** запуск *SimpleGenerator* и *SimpleIntegrator* в отдельных потоках; обмен задачами через блокирующие методы *Task*.
- **Проверка:** отсутствие *NullPointerException*, соответствие пар *Source/Result*, корректное завершение потоков.

```

public void putNoSync(Function function, double left, double right, double step) {
    this.function = function;
    this.left = left;
    this.right = right;
    this.step = step;
}

public TaskSnapshot takeNoSync() {
    return new TaskSnapshot(function, left, right, step);
}

public static class TaskSnapshot {
    public final Function function;
    public final double left;
    public final double right;
    public final double step;

    public TaskSnapshot(Function function, double left, double right, double step) {
        this.function = function;
        this.left = left;
        this.right = right;
        this.step = step;
    }
}

```

- **Задание 4** — *complicatedThreads()*
 - **Реализация:** запуск *Generator/Integrator* (потомков *Thread*), синхронизация через *SimpleSemaphore*, обмен через неблокирующие методы *Task*. Основной поток ждёт 50 ms и затем вызывает *interrupt()*; потоки корректно обрабатывают прерывание.
 - **Проверка:** все сгенерированные задания корректно обрабатываются; прерывания не оставляют незавершённых задач.

```

@Override
public void run() {
    int count = task.getCount();
    for (int i = 0; i < count; i++) {
        double base = 1.0 + rnd.nextDouble() * 9.0;
        Function f = new Log(base);
        double left = rnd.nextDouble() * 100.0;
        double right = 100.0 + rnd.nextDouble() * 100.0;
        double step = rnd.nextDouble();
        try {
            task.putBlocking(f, left, right, step);
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
            break;
        }
        System.out.printf(format: "Source %.6f %.6f %.6f\n", left, right, step);
    }
}

```

4. Примеры вывода :

=== Демонстрация лабораторной работы №6 ===

Exact integral of exp on [0,1]: 1.718281828459

Found step 0.000488281250000 gives value 1.718281862598 (err=3.41391943603e-08)

--- nonThread() ---

```

Source 67.019715 197.225205 0.169691
Result 67.019715 197.225205 0.169691 329.699471
Source 73.520643 166.083619 0.966672
Result 73.520643 166.083619 0.966672 743.161084
Source 36.177011 192.379452 0.515950
Result 36.177011 192.379452 0.515950 377.113016
Source 81.769195 129.431592 0.027341
Result 81.769195 129.431592 0.027341 203.386082
Source 54.552859 123.093158 0.691155
Result 54.552859 123.093158 0.691155 161.250614
Source 31.852533 187.390621 0.148545
Result 31.852533 187.390621 0.148545 345.679485
Source 20.057656 152.744523 0.260912
Result 20.057656 152.744523 0.260912 460.716950
Source 87.274158 101.154691 0.483529
Result 87.274158 101.154691 0.483529 68.423940
Source 13.200995 105.955568 0.032423
Result 13.200995 105.955568 0.032423 178.649454
Source 14.242562 175.094871 0.948513
Result 14.242562 175.094871 0.948513 393.649725
Source 32.434178 156.361543 0.109348
Result 32.434178 156.361543 0.109348 357.965646
Source 60.449401 100.586465 0.216302
Result 60.449401 100.586465 0.216302 77.522403

```

Source 11.774197 194.562240 0.793880
Result 11.774197 194.562240 0.793880 406.190157
Source 35.074471 169.275855 0.277821
Result 35.074471 169.275855 0.277821 420.682026
Source 14.849866 161.020783 0.339978
Result 14.849866 161.020783 0.339978 303.023857
Source 96.658117 153.052686 0.700501
Result 96.658117 153.052686 0.700501 1347.937315
Source 24.510674 153.675583 0.811620
Result 24.510674 153.675583 0.811620 249.839391
Source 80.481704 172.992969 0.962391
Result 80.481704 172.992969 0.962391 215.846392
Source 72.139011 145.031032 0.550016
Result 72.139011 145.031032 0.550016 298.170823
Source 22.096765 104.538789 0.490941
Result 22.096765 104.538789 0.490941 708.594480
Source 75.806402 168.492596 0.003672
Result 75.806402 168.492596 0.003672 239.778194
Source 58.643444 191.180177 0.285328
Result 58.643444 191.180177 0.285328 614.251294
Source 81.414336 158.711910 0.925214
Result 81.414336 158.711910 0.925214 213.966223
Source 14.598900 163.904838 0.251251
Result 14.598900 163.904838 0.251251 314.826753
Source 25.960358 155.707499 0.188878
Result 25.960358 155.707499 0.188878 567.110190
Source 3.596243 188.681290 0.627094
Result 3.596243 188.681290 0.627094 653.212340
Source 67.070035 107.526791 0.732574
Result 67.070035 107.526791 0.732574 139.233433
Source 2.954649 189.241112 0.130366
Result 2.954649 189.241112 0.130366 383.613209
Source 47.935385 131.214969 0.378427
Result 47.935385 131.214969 0.378427 380.485014
Source 82.561308 117.425860 0.703324
Result 82.561308 117.425860 0.703324 82.364308
Source 91.800466 116.856375 0.535706
Result 91.800466 116.856375 0.535706 679.354519
Source 94.745771 168.364324 0.986681
Result 94.745771 168.364324 0.986681 168.209094
Source 45.134680 184.454022 0.040043
Result 45.134680 184.454022 0.040043 633.597405
Source 56.787383 162.745723 0.900312
Result 56.787383 162.745723 0.900312 392.025421
Source 13.634447 128.412871 0.632850
Result 13.634447 128.412871 0.632850 212.187120
Source 82.696375 143.454466 0.959858
Result 82.696375 143.454466 0.959858 135.751990

Source 60.193901 109.392506 0.834838
Result 60.193901 109.392506 0.834838 96.364687
Source 45.909279 158.832136 0.208143
Result 45.909279 158.832136 0.208143 239.397743
Source 58.985174 178.294796 0.217550
Result 58.985174 178.294796 0.217550 263.176858
Source 79.539259 129.832264 0.890292
Result 79.539259 129.832264 0.890292 253.678335
Source 24.788928 142.152221 0.326809
Result 24.788928 142.152221 0.326809 752.338881
Source 45.822016 123.333571 0.251487
Result 45.822016 123.333571 0.251487 160.733167
Source 61.715782 106.682423 0.761985
Result 61.715782 106.682423 0.761985 159.294189
Source 31.397858 178.877278 0.086229
Result 31.397858 178.877278 0.086229 438.586956
Source 0.459203 155.274673 0.165001
Result 0.459203 155.274673 0.165001 327.265148
Source 50.070591 161.094109 0.918048
Result 50.070591 161.094109 0.918048 246.764892
Source 18.220526 149.173829 0.122958
Result 18.220526 149.173829 0.122958 504.108248
Source 59.683834 135.086618 0.938543
Result 59.683834 135.086618 0.938543 196.475207
Source 60.446729 158.627439 0.238321
Result 60.446729 158.627439 0.238321 372.453158
Source 24.755382 187.032467 0.084668
Result 24.755382 187.032467 0.084668 2232.178306
Source 8.521565 159.761593 0.870537
Result 8.521565 159.761593 0.870537 559.997115
Source 81.510278 137.274275 0.883170
Result 81.510278 137.274275 0.883170 117.914352
Source 62.550689 156.951170 0.303013
Result 62.550689 156.951170 0.303013 374.755537
Source 86.953157 118.161659 0.941086
Result 86.953157 118.161659 0.941086 65.360216
Source 33.381413 115.450087 0.339357
Result 33.381413 115.450087 0.339357 315.903075
Source 58.620548 113.911100 0.211196
Result 58.620548 113.911100 0.211196 107.271088
Source 62.997683 114.047201 0.582780
Result 62.997683 114.047201 0.582780 161.401503
Source 36.653724 184.749266 0.411358
Result 36.653724 184.749266 0.411358 309.074609
Source 17.430964 152.513432 0.902576
Result 17.430964 152.513432 0.902576 323.791826
Source 24.832726 122.605387 0.226603
Result 24.832726 122.605387 0.226603 284.406716

Source 21.327582 173.605817 0.841575
Result 21.327582 173.605817 0.841575 696.407147
Source 92.212209 110.618348 0.172930
Result 92.212209 110.618348 0.172930 38.919536
Source 28.313199 184.394038 0.968488
Result 28.313199 184.394038 0.968488 322.569052
Source 41.999701 183.724088 0.251671
Result 41.999701 183.724088 0.251671 316.673021
Source 46.295032 186.675110 0.429038
Result 46.295032 186.675110 0.429038 342.709990
Source 79.316939 112.588963 0.773368
Result 79.316939 112.588963 0.773368 66.143377
Source 61.842455 151.241804 0.199792
Result 61.842455 151.241804 0.199792 343.554440
Source 34.484872 158.154658 0.980035
Result 34.484872 158.154658 0.980035 337.687358
Source 32.786561 195.770480 0.798594
Result 32.786561 195.770480 0.798594 361.519553
Source 23.472971 169.060325 0.344742
Result 23.472971 169.060325 0.344742 376.006043
Source 43.179381 151.093914 0.455514
Result 43.179381 151.093914 0.455514 222.912884
Source 52.153488 119.207519 0.677398
Result 52.153488 119.207519 0.677398 205.472220
Source 39.753316 125.624762 0.998206
Result 39.753316 125.624762 0.998206 186.702528
Source 96.001239 107.733588 0.415669
Result 96.001239 107.733588 0.415669 28.243717
Source 94.021466 102.695287 0.955593
Result 94.021466 102.695287 0.955593 31.440380
Source 11.879604 168.659149 0.729906
Result 11.879604 168.659149 0.729906 333.640334
Source 59.159430 130.578967 0.633783
Result 59.159430 130.578967 0.633783 1939.239443
Source 63.553312 162.475237 0.321015
Result 63.553312 162.475237 0.321015 1674.370059
Source 15.809235 189.959619 0.168114
Result 15.809235 189.959619 0.168114 932.097056
Source 67.568665 146.935802 0.040167
Result 67.568665 146.935802 0.040167 187.239108
Source 25.946016 148.613655 0.805249
Result 25.946016 148.613655 0.805249 585.192975
Source 91.141808 149.321128 0.709244
Result 91.141808 149.321128 0.709244 192.962675
Source 38.948042 138.071085 0.919207
Result 38.948042 138.071085 0.919207 292.449377
Source 58.695707 187.437846 0.770942
Result 58.695707 187.437846 0.770942 269.112463

Source 19.010770 112.020933 0.515262
Result 19.010770 112.020933 0.515262 242.059933
Source 48.686397 111.621131 0.647432
Result 48.686397 111.621131 0.647432 149.249648
Source 88.045491 171.226635 0.225591
Result 88.045491 171.226635 0.225591 1755.509176
Source 87.359199 165.004728 0.936479
Result 87.359199 165.004728 0.936479 178.419454
Source 3.203228 164.717081 0.050594
Result 3.203228 164.717081 0.050594 326.449497
Source 71.075457 184.324418 0.823935
Result 71.075457 184.324418 0.823935 381.578514
Source 13.164503 102.018934 0.349785
Result 13.164503 102.018934 0.349785 266.619328
Source 73.850493 188.520567 0.341509
Result 73.850493 188.520567 0.341509 251.575972
Source 62.335605 136.195795 0.836163
Result 62.335605 136.195795 0.836163 245.575380
Source 21.743342 127.405358 0.348712
Result 21.743342 127.405358 0.348712 196.968389
Source 37.091973 102.459922 0.577968
Result 37.091973 102.459922 0.577968 160.983792
Source 40.738922 187.660367 0.367899
Result 40.738922 187.660367 0.367899 370.918725
Source 73.209311 122.122053 0.261727
Result 73.209311 122.122053 0.261727 106.887333
Source 9.763278 178.430054 0.651302
Result 9.763278 178.430054 0.651302 440.836563
Source 68.623091 106.645001 0.220500
Result 68.623091 106.645001 0.220500 231.789865
Source 75.472484 143.776862 0.108960
Result 75.472484 143.776862 0.108960 442.418611

--- simpleThreads() ---

Source 0.239781 199.873322 0.405251
Source 85.227984 146.899429 0.592921
Result 0.239781 199.873322 0.405251 403.827237
Result 85.227984 146.899429 0.592921 267.810092
Source 67.926801 141.326736 0.573658
Result 67.926801 141.326736 0.573658 174.267871
Source 84.578481 177.747933 0.164130
Result 84.578481 177.747933 0.164130 409.097023
Source 38.068113 187.325524 0.835119
Result 38.068113 187.325524 0.835119 543.025549
Source 65.170303 168.939969 0.496595
Result 65.170303 168.939969 0.496595 255.237723
Source 12.341638 194.841987 0.819322
Source 22.210412 124.527916 0.700918

Result 12.341638 194.841987 0.819322 366.605574
Result 22.210412 124.527916 0.700918 211.246607
Source 75.981515 132.529693 0.555609
Source 33.165799 105.908543 0.209455
Result 75.981515 132.529693 0.555609 417.781212
Source 59.359609 195.464340 0.316583
Result 33.165799 105.908543 0.209455 158.029662
Source 35.241605 111.020031 0.377045
Result 59.359609 195.464340 0.316583 324.817930
Result 35.241605 111.020031 0.377045 161.584849
Source 95.520402 135.613349 0.633175
Result 95.520402 135.613349 0.633175 86.132488
Source 55.257926 196.025576 0.348879
Result 55.257926 196.025576 0.348879 406.682500
Source 42.809867 167.799863 0.235058
Source 73.565916 111.749185 0.074437
Result 42.809867 167.799863 0.235058 258.468845
Source 66.738970 135.926754 0.920321
Result 73.565916 111.749185 0.074437 639.132208
Result 66.738970 135.926754 0.920321 249.067705
Source 28.643839 152.129288 0.596379
Result 28.643839 152.129288 0.596379 282.838798
Source 0.222412 149.226251 0.608312
Result 0.222412 149.226251 0.608312 294.109374
Source 94.013664 136.955804 0.993318
Result 94.013664 136.955804 0.993318 111.967876
Source 26.333244 123.300851 0.079556
Result 26.333244 123.300851 0.079556 217.940629
Source 84.118850 176.156280 0.605350
Result 84.118850 176.156280 0.605350 199.467597
Source 52.713963 185.721082 0.156643
Result 52.713963 185.721082 0.156643 120027.827201
Source 70.803925 187.896722 0.421287
Result 70.803925 187.896722 0.421287 268.828361
Source 71.784757 130.945812 0.330011
Result 71.784757 130.945812 0.330011 145.586126
Source 17.036519 167.762753 0.756464
Result 17.036519 167.762753 0.756464 296.309094
Source 25.889448 134.330673 0.082518
Result 25.889448 134.330673 0.082518 439.201799
Source 59.512424 131.286713 0.600047
Result 59.512424 131.286713 0.600047 166.500841
Source 21.519854 159.990235 0.416706
Result 21.519854 159.990235 0.416706 683.585309
Source 96.784358 177.086341 0.260738
Result 96.784358 177.086341 0.260738 342.405504
Source 61.759107 122.279595 0.338676
Result 61.759107 122.279595 0.338676 242.502700

Source 64.555038 142.880644 0.189791
Result 64.555038 142.880644 0.189791 459.027949
Source 91.084615 185.660944 0.302473
Source 4.775161 151.648589 0.110240
Result 91.084615 185.660944 0.302473 206.637612
Source 71.671246 112.720787 0.868879
Result 4.775161 151.648589 0.110240 365.890916
Result 71.671246 112.720787 0.868879 82.563965
Source 60.186967 170.196254 0.831243
Result 60.186967 170.196254 0.831243 262.611724
Source 35.335409 195.103424 0.082540
Result 35.335409 195.103424 0.082540 637.041080
Source 50.953967 127.398034 0.376461
Result 50.953967 127.398034 0.376461 192.905786
Source 80.260594 127.126301 0.321317
Result 80.260594 127.126301 0.321317 148.105247
Source 15.017488 164.377559 0.305358
Result 15.017488 164.377559 0.305358 336.476722
Source 6.645073 106.842888 0.621768
Result 6.645073 106.842888 0.621768 211.023722
Source 18.820639 113.893925 0.356346
Result 18.820639 113.893925 0.356346 173.987944
Source 10.412294 174.318679 0.734681
Result 10.412294 174.318679 0.734681 323.108877
Source 70.040351 131.198876 0.790675
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